Model Card — Virtue-Based Reflective Scaffolding Agent (planned, v0.1)

Deliverable 07 · Research Methods and Scientific Integrity in AI · UNMSM 2026-II · Deisy Lizbeth Acosta Ticse Framework: Mitchell et al. (2019), *Model Cards for Model Reporting*, FAT* '19, 220–229. https://doi.org/10.1145/3287560.3287596

Adaptation note. Model cards were proposed for *trained models*. This artifact is a **system built on a base large language model** whose behaviour is set by prompting and, possibly, retrieval; no model is trained in the planned design. The card therefore documents the system, and refers base-model properties to the provider's own documentation. **Status: PLANNED — NOT YET IMPLEMENTED OR EVALUATED.**

1. Model details

Element	Documentation	Status
Type	LLM-based conversational agent implementing Socratic, virtue-oriented scaffolding when a student asks it to complete academic work	Defined
Base model	Local open model vs commercial API not yet chosen; the choice depends on privacy (no student data to third-party providers without a data-processing agreement)	UNKNOWN
Configuration	System prompt and scaffolding policy derived from the annotated corpus (see datasheet)	Planned
Research approach	Design Science Research; contribution: nascent design theory (Hevner et al., 2004; Gregor & Hevner, 2013)	Defined

2. Intended use

Element	Documentation	Status
Primary use	Prompt undergraduates to deliberate (prudence, honesty) before delegating academic work to generative AI	Defined

Element	Documentation	Status
Users	Undergraduates in Peruvian university courses, in Spanish	Defined
Out of scope	Grading; misconduct detection or sanctioning; replacing the instructor; use outside the evaluated courses without re-evaluation	Defined

3. Factors

Element	Documentation	Status
Relevant factors	Course section (unit of assignment), discipline, prior AI use, baseline reflection level	Planned
Evaluation factors	Results reported per condition and, where sample allows, per section	Planned

4. Metrics

Element	Documentation	Status
Metacognitive reflection	Coding of reflections/logs with the four-category scheme of Kember et al. (2008)	Defined (Spanish adaptation pending)
AI dependence	DAI scale (Morales-García et al., 2024), developed with Peruvian university students	Defined
Product integrity	Composite: log-based provenance ratio + blinded expert rubric	Planned (content validity pending)
Manipulation check	Perceived restrictiveness of the restrictive-normative condition	Planned
Behavioural check	Rate at which the agent withholds full solutions (answer leakage)	Planned

5. Evaluation data

Element	Documentation	Status
Design	Sequential explanatory mixed methods; quasi-	Defined

Element	Documentation	Status
	experiment with three conditions (agent, unrestricted, restrictive-normative); intact section as unit, ≥ 2 sections per condition; multilevel models	
Data	Pre/post measures and conversation logs — **NOT YET COLLECTED**	UNKNOWN
Lesson from Deliverable 04	Include an unassisted post-test, since gains with the tool may not persist without it	Planned

6. Training data

No model is trained. The annotated corpus informs the system prompt and scaffolding policy (see datasheet). If fine-tuning is later adopted, this section must be completed. — **Not applicable (planned design)**

7. Quantitative analyses

None yet. Results will be reported for every outcome, by condition, with effect sizes, whether or not they support the hypotheses. — **UNKNOWN**

8. Ethical considerations

Risk	Mitigation	Status
Researcher is designer, instructor and evaluator	Ethics approval; consent collected by a third party and decoupled from grading; blinded external coders	Planned
Student data exposure	Logs pseudonymised at capture; institution-controlled storage; never in public repositories	Defined
Paternalism / frustration	Agent asks rather than refuses; monitor disengagement in logs	Planned
Moralising or cultural bias in virtue judgments	Exemplar-anchored annotation; expert validation	Planned

9. Caveats and recommendations

Evidence from the Deliverable 04 review shows that tool-level scaffolding improved process measures more consistently than unassisted performance; claims about the agent should be

limited to what the evaluation measures. The agent must not be presented as a detector of academic misconduct.

References

Gregor, S., & Hevner, A. R. (2013). *MIS Quarterly, 37*(2), 337–355. https://doi.org/10.25300/MISQ/2013/37.2.01 · Hevner, A. R., et al. (2004). *MIS Quarterly, 28*(1), 75–105. https://doi.org/10.2307/25148625 · Kember, D., et al. (2008). *Assessment & Evaluation in Higher Education, 33*(4), 369–379. https://doi.org/10.1080/02602930701293355 · Morales-García, W. C., et al. (2024). *Frontiers in Education, 9*, 1323898. https://doi.org/10.3389/feduc.2024.1323898